

Data Structures & Algorithms

BCA — Semester III — 2025 — Answer Sheet
Subject Code: CACS201

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Group B

2. Explain stack as ADT. Describe push and pop operation in stack.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. Define linear and circular queue. Write are the limitation of linear queue? How circular queue can used to overcome the limitation of linear queue.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

4. What is hashing? Explain different types of collision resolution techniques with suitable example.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. What is recursion? Write an algorithm to solve Tower of Hanoi problem.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. Differentiate between internal sorting and external sorting algorithm. Explain insertion sorting algorithm with example.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. How graph can be represented using adjacency matrix? Explain Depth first traversal in a graph with suitable example.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. A binary tree T has 9 nodes. The inorder and preorder traversals of T yield the following sequence of nodes: In-order: R Z J T K H N M P Pre-order: K Z R T J N H P M Draw the tree T. OR Write short notes on: a. Deterministic and non-deterministic algorithm b. Divide and conquer algorithm

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

Group C

Attempt any TWO questions[2x10=20]

9. Describe types of linked list. Write an algorithm to perform following operation in circular linked list:• Insert node at the beginning of circular linked list.• Insert node at the end of circular linked list.• Delete node from the beginning of circular linked list.• Delete node from the end of circular linked list.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. Differentiate between strict binary tree and skewed tree. Explain binary search tree and insertion and deletion operation in BST in detail. Consider following data and construct BST: 14, 11, 12, 19, 15, 22, 13, 8, 33, 7, 9, 20.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. What is priority queue? Explain sequential searching algorithm and binary searching algorithm with suitable illustration.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

These answer sheets are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.